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Identifying Transmission Opportunity through Transmission Power and Bit Rate for Improved VANET Efficiency

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Abstract
With the development of vehicular industry and wireless communication technology, vehicular ad hoc network (VANET) has become a popular research field. However, timely dissemination of safety related information in VANET is always challenging due to the highly dynamic vehicular environment. In this paper, we investigate an analytic approach in enhancing transmission efficiency of VANET by adapting transmission power and bit rate for IEEE 802.11p based VANET. The proposed algorithm, Joint Adaptation of Transmission power and Bit rate (JATB) algorithm uses modified Dynamic Programming (DP) to search for optimal value of transmission power and bit rate with the aim to increase packet success rate, minimize end-to-end delay and minimize busy time. JATB also uses vehicle density to improve vehicular environment awareness, and adapt transmission power and bit rate accordingly. Adaptability of JATB algorithm is tested with different beacon rates and simulation scenarios. Simulation results show an improvement in transmission efficiency as compared to standard protocol [original algorithm set in VANET standard which has fixed transmission power (20 mW) and fixed bit rate (6Mbps)] and reference journal's Transmission Power Transmission Range (TPTR) [algorithm from reference journal that adapts transmission power according to transmission range] algorithms.

Keywords Vehicular ad-hoc network · IEEE 802.11p · IEEE 1609 · Transmission efficiency · Dynamic programming

1 Introduction

In recent years, number of vehicles on the road has increased exponentially. This phenomenon has caused high density of traffic in already dense city, which in turn causes accidents and congestions. There is a dire need to improve the traffic management and ease congestion. Several studies have pointed out that 60% accidents could be avoided if warning messages were provided to the drivers before the accident. [1] Therefore, researchers have introduced Intelligent Transportation System (ITS) to advance car technology towards sustainable transportation system. This is where Vehicular Ad-hoc Network (VANET), a part of ITS steps in, by allowing communication between vehicles to take place to share information in the form of messages or images on road hazards, and traffics or accidents to other vehicles.

VANET is an emerging and promising technology that enables vehicles to communicate with their neighboring vehicles even in the absence of central base station. The communication between vehicles (nodes) happen through multi hops across intermediate nodes to relay information beyond the reach of radio transmission. The main characteristic of VANET is its infrastructure-less and this enables VANET to be implemented without incurring a huge amount of cost to the government, therefore increasing VANET desirability and its commercial value.

Wireless Access in Vehicular Network (WAVE) [2] is designed to support ITS on 5.9GHz frequency band and to adhere to IEEE 802.11p with IEEE 1609 standard family [3–6]. VANET uses Dedicated Short-Range Communication (DSRC) technique for enhancement of transportation system. US Federal Communication Commission has reserved 75 MHz of DSRC spectrum (5.85–5.925GHz) to be used for Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication.

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Potential of VANET is wide and several researchers have evaluated and listed potential VANET applications through different papers. Two categories have been formed from the extensive potential VANET applications, they are safety-related and non-safety-related applications.

Safety-related-applications are further classified into three basic categories, namely: driver assistance, alert information and warning alert. Safety messages from the above categories require stringent delay to prevent the safety information from becoming obsolete. Furthermore, these safety messages are broadcast oriented, therefore it must have deep penetration across the entire network and deliver reliably to the intended recipients. Non-safety-related applications in VANET are commercial applications that aim to improve traffic efficiency and passenger comfortability.

Many researchers have undergone extensive research to improve VANET transmission efficiency. Some of the parameters that are widely used are transmission power, bit rate and contention window. The modification of these parameters has brought improvement in VANET transmission efficiency but do come with their own shortcomings. Sec.2 discusses related work from prestigious researchers that uses transmission power and bit rate to enhance transmission efficiency of VANET.

This paper aims to improve safety-related applications of VANET through the enhancement of beacon transmission which is used to update neighboring vehicles with information regarding vehicles' density, speed, position or their headings. Constant beacon transmission allows drivers to be aware of its environment in real time. It also allows the implementation of driver assistance, alert information and warning alert. In summary, enhancement of VANET beacon transmission will be done through increasing packet success rate and reducing end-to-end delay and busy time through manipulation of MAC layer and protocol parameters of IEEE 802.11p IEEE 1609 standards.

The contributions of this paper are as follow: Firstly, transmission power and bit rate are proposed to be used as the parameters to determine transmission opportunity. Secondly, an algorithm with joint adaptation of transmission power and bit rate is proposed to improve VANET transmission efficiency. Thirdly, the proposed algorithm is verified through real time simulation in urban map of Malaysia to evaluate the performance of the proposed algorithm against standard protocol [7] and TPTR algorithm [8]. The results show that the proposed algorithm improved the packet success rate, end-to-end delay and busy time as compared to the standard protocol [7] and TPTR algorithm [8].

The rest of the paper is organized as follows. Section 2 illustrates the existing work. Section 3 discusses the proposed JATB algorithm. Section 4 presents the output parameters which would be used to evaluate algorithm performance. Section 5 presents the algorithm from existing work for comparison and discusses the simulation, analytical parameters and result. Section 6 concludes the paper.

2 Related work

Several researchers have explored the possibility of improving VANET transmission efficiency through adapting transmission power. In [9], Kim ea. al. and his team proposed a beacon update scheme that decreases transmission delay. The beacon update scheme uses Transmission Power Control (TPC) to adjust transmission range whereas Bi-Beacon scheme transmits two types of beacon to cover both long and short range. This scheme considers more parameter to determine vehicular environment instead of only vehicle speed. So, it motivates me to explore a scheme that uses less parameter to determine vehicular environment accurately. In [10], Sofiane ea. al. has proposed a scheme that adapts beacon transmission rate and power using key concepts of binary search algorithm to search for optimal busy ratio. This scheme allows faster reaction and adaptation to channel condition. However, binary search algorithm requires large amount of time, therefore a more efficient optimization algorithm that uses less time need to be determined.

Danda ea. al. proposed a scheme in [8] to ensure low delay in beacon transmission. This scheme uses periodically beacon transmission to estimate vehicle density. This information is used to adapt transmission power with current transmission environment. High throughput and low delay can be achieved through this scheme. However, it uses high transmission power which would cause high interference to neighboring node, therefore a reasonable transmission power value need to be considered.

Besides transmission power, bit rate has also been explored extensively by researchers. In [11], Yuan ea. al. proposed Density-Aware Rate Adaptation (DARA) that considers interference resembling real network environment without additional frame. This scheme obtains information required to estimate the average packet loss rate (PLR_{avg}). The average packet loss rate is used for bit rate selection. However, without the support of control frame, severe interference could occur. Statistic-based rate adaptation algorithm is proposed in [12]. This algorithm estimates the current channel qualities and determines the best bit rate for outgoing frame using statistic. It uses history information of location, velocity and vehicle density from database and combines it with current transmission information. Historical statistics are obtained in global database (Road side unit, RSU) and local storage (vehicle). This proposed algorithm maximizes network throughput but the need of RSU limits the potential of VANET and it requires large resources to implement, so a better algorithm that does not incur additional infrastructure would be better. Since majority rate adaptation schemes are designed for indoor

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environment, which are characterized by slow channel changes, thus they are not suitable for vehicular communication which functions in highly dynamic topology.

Therefore, [13] proposed Random Forest Rate Adaptation (RFRA). Random Forests algorithm is used to learn the propagation environment and select suitable data rate for transmission. The selection of suitable data rate is done based on configuration file which is generated through implementation of Random Forests (RF) algorithm on training data. However, the use of Random Forest algorithm requires RSU and large amount of data to improve accuracy, therefore this journal has motivated me to search for better algorithm which is not resource intensive.

Through review of related work, we discovered that joint adaptation of transmission power and bit rate is not commonly used.

Furthermore, the use of vehicle density and channel condition to improve the awareness of the vehicle in adapting protocol parameter is widely adopted by researchers. It allows protocol parameters to utilize information from channel condition and vehicle environment to adapt to the transmission environment. The increase of vehicle awareness has demonstrated its benefits through various researches as the data transmission efficiency has shown significant improvement.

The use of optimization method is an effective way in choosing optimal value for transmission power and bit rate. From the literatures, it can be observed that parameters such as transmission power and bit rate have not been incorporated into vehicular environment awareness with the use of optimization method. This method is explored in our proposed algorithm to further enhance the transmission efficiency.

3 Proposed JATB algorithm

From numerous literatures in Sec. 2, the effectiveness of manipulating transmission power and bit rate in improving transmission efficiency is evident. Therefore, the potential of joint adaptation of both transmission power and bit rate is huge in improving transmission efficiency even further. To the best of my knowledge, scheme for joint adaptation of transmission power and bit rate is limited in this field Fig. 1.

High transmission power causes interference. Low transmission power causes packets to be dropped. Therefore, it is important to control the transmission power to ensure the power is at moderate level to ensure packets are successfully sent without interference, and with adequate power. High bit rate causes congestion in the network, whereas low bit rate causes inefficient channel usage. The bit rate should be adaptable according to channel congestion condition. Hence, a joint adaptation of transmission power and bit rate have been selected as controlled parameter in this paper. Figure. 2a shows packets dropped even with a high bit rate due to low transmission power. Figure. 2b shows packets dropped due to interference and low bit rate (network congestion caused by high volume of vehicles). Therefore, it is important to jointly adapt transmission power and bit rate as they are dependent on each other.

Transmission power is the determinant of the transmission range in vehicular communication, whereas bit rate determines the transmission speed of data across network. In the standard protocol [7], both these parameters are kept constant. This is not an ideal set up for VANET as VANET operates in highly dynamic environment that deals with high mobility and high density of vehicles. Therefore, transmission power and bit rate need to have the ability to adapt to the dynamic environment. Hence, in the following proposed methodology, an adaptive transmission power and bit rate algorithm is proposed.

The flow chart of the formation of Proposed JATB algorithm is shown in Fig. 1. Figure 3 shows the pseudo code that will be inserted into OMNeT++ Mac1609_4 layer to enable the node to access the lookup table that has been constructed

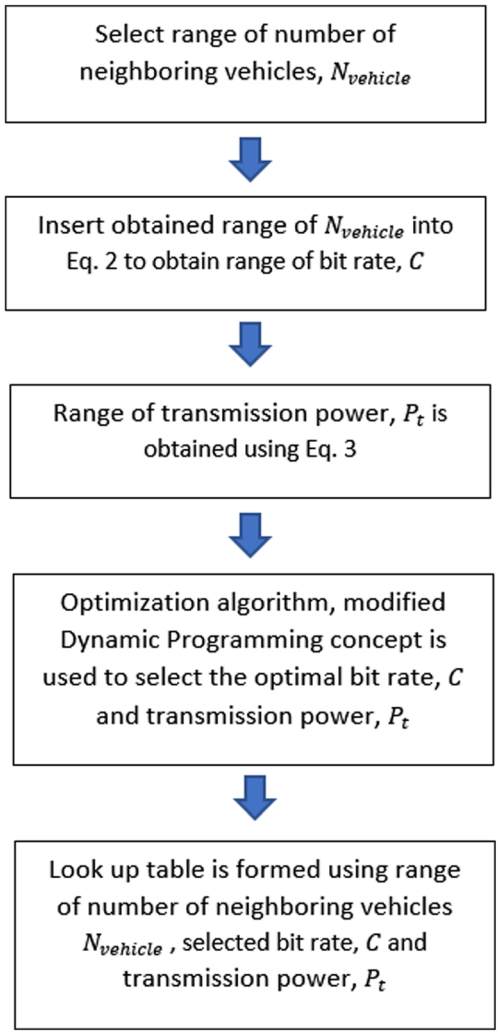

Fig. 1 Flow chart of the formation of proposed JATB algorithm

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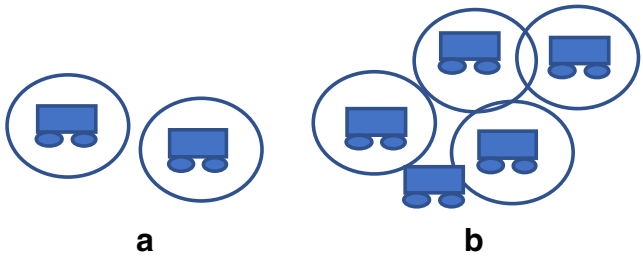

Fig. 2 a: Non-congested channel: No overlapping interference regions with high bit rate. b: Congested channel: Overlapping interference regions with low bit rate

using proposed JATB algorithm and adapt its transmission power and bit rate according to number of neighboring vehicles.

3.1 Vehicle density adaptation

The need for algorithm to adapt with its environment is driven by the dynamically changing vehicular environment and channel condition. Therefore, Eq. 1 from [14] is used to compute number of neighboring vehicles. Eq. 1 is used to estimate vehicle density because it uses received packets and number of packet collisions to predict the traffic and channel conditions.

$$N_{vehicle} = \frac{(n_{i_{recv}} + n_{i_{dec}} + 2*n_{i_{coll}})}{Rate_{avg}} \tag{1}$$

where $N_{vehicle}$ is the estimated number of neighboring vehicles, $n_{i_{recv}}$ is the number of beacons received during period i, $Rate_{avg}$ is the average beacon rate of neighbors, $n_{i_{dec}}$ is the number of collisions due to non-decodable beacons from senders in interference range, $n_{i_{coll}}$ is the number of collisions due to beacon collisions of two senders both in the range of receiver.

3.2 Bit rate adaptation

Bit rate is calculated based on Eq. 2. This equation is derived from [14]. It uses $N_{vehicle}$ calculated from Eq. 2. This allows bit rate calculation to consider its vehicular environment. Equation 2 uses fixed beacon size which of 500 bytes, and beacon rate is to 0.05 s, 0.04 s and 0.03 s based on simulation scenarios.

$$C = \frac{b_{frequency} * b_{size} * N_{vehicle}}{\beta} \tag{2}$$

where C is the bit rate in bits per second, $b_{frequency}$ is beacon frequency in Hertz (Hz), b_{size} is beacon size, $N_{vehicle}$ is the number of neighboring vehicles which is obtained from Eq. 1, and β is a constant set to 0.17. [14]

```
JATB Algorithm Joint Adaptation of Transmission power
and Bit rate

// N_vehicle : Number of neighboring vehicles

// txPower: Transmission Power in mW

// bitrate: bit rate in bits per second

//A, B: Number of neighboring vehicle

//C: Transmission power 1 (mW)

//D: Bit rate 1 (Mbps)

//E: Transmission power 2 (mW)

//F: Bit rate 2 (Mbps)

// n_i_recv: number of packets received during period i

// n_i_dec: number of collisions due to non-decodable packet

// n_i_coll: number of collisions due to packet collision from
two senders both in range of receiver

Input: n_i_recv, n_i_dec, n_i_coll, Rate_avg

Output: txpower, bitrate

Step 1: N_vehicle= (n_i_recv + n_i_dec + 2*n_i_coll ) /
(Rate_avg)

Step 2: switch (N_vehicle){

        case A … B:

        txPower = C;

        bitrate = D;

        break;

        default:

        txPower = E

        bitrate = F;

        break;}
```

Fig. 3 Proposed JATB algorithm pseudo code

3.3 Transmission power

Friis Transmission Formula in Eq. 3 is used to calculate transmission power.

$$P_t = \frac{P_r d^2 (4\pi^2) L}{G_t G_r \lambda^2} \tag{3}$$

where P_t is the transmission power in milliwatt (mW), P_r is the received power in milliwatt (mW), d is the distance between transmitter and receiver in meter (m), L is the system loss, $G_t$is the transmitter antenna gain and G_r is the receiver antenna gain, λ is wavelength [15].

Optimization algorithm is used to obtain the range of transmission power and bit rate. DP is a powerful algorithm design technique that is used to solve complex problem by breaking the problems into collection of sub problems. These sub problems are optimized with the objective of searching for the best optimal value. In this paper, DP concept is used and its sub-problems are transmission power and bit rate calculations.

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Both these subproblems are then used to calculate the main equation using Shannon-Hartley theorem shown in Eq. 4 which is further derived with DP optimization

$$C = B\log_2\left(1 + \frac{S}{N}\right) \tag{4}$$

where C is bitrate in bits per second, B is the bandwidth of the channel in Hertz, S is transmission power over bandwidth in milliwatt (mW), and N is the average of interference plus noise power over bandwidth.

Shannon-Hartley theorem relates transmission power, noise and interference. Original DP has high complexity and needs to undergo exhaustive search for all possible solutions which is inefficient. Therefore, instead of searching for all possible solutions, modified DP search through a range of possible solutions defined by a range, D. However, the range D is limited by d and $N_{vehicle}$ as expressed in Eqs. 5 and 6. Eqs. 5 and 6 are used to express the working principle of modified DP that obtain optimal transmission power, $\widehat{P}_t$ and optimal bit rate, $\hat{C}$ by searching through a range of transmission power, P_t and bit rate, C. Range of transmission power value is obtained from Eq. 3 and threshold by d. As for bit rate, the range of bit rate is obtained from Eq. 2 and threshold by $N_{vehicle}$.

$$\widehat{P}_t = \max_{d \in DP_t(min,max)} \{Packet_{SR}\} \quad \min_{d \in DP_t(min,max)} \{EEdelay\} \quad \min_{d \in DP_t(min,max)} \{BusyTime\} \tag{5}$$

$$\hat{C} = \max_{N_{vehicle} \in DC(min,max)} \{Packet_{SR}\} \quad \min_{N_{vehicle} \in DC(min,max)} \{EEdelay\} \quad \min_{N_{vehicle} \in DC(min,max)} \{BusyTime\} \tag{6}$$

Table 1 Lookup table of reference journal

Number of SNIR Lost Packet	Transmission Power (mW)
1–2	20
3–5	19
6–8	18
9–11	17
12–14	16
15–17	15
18–20	14
21–23	13
24–26	12
27–29	11
> 29	10

Table 2 Proposed JATB algorithm transmission power range and bit rate range

$N_{vehicle}$	Transmission Power (mW)	Bit Rate (Mbps)
0–1	18–20	1.8
2–6	14–17	3.8–11.3
7–9	11–15	13.2–16.9
10–13	8–12	18.8–24.5
> 13	2–5	> 24.5

In Eq. 5 and Eq. 6, $\widehat{P}_t$ is defined as the optimal transmission power in milliwatt (mW). P_t is defined as the transmission power in milliwatt (mW) where d is the distance between transmitter and receiver in meter (m). D is defined as the range whereas $Packet_{SR}$ is defined as the Packet Success Rate.$EEdelay$ is the End-to-End delay, $N_{vehicle}$ is the number of neighboring vehicles, $\hat{C}$ is the optimal bit rate (Mbps) and C is the bit rate (Mbps).

With $N_{vehicle}$ and range of transmission power and bit rate obtained, the proposed JATB algorithm can be formed. Optimal transmission power and bit rate are obtained through extensive simulation to identify most optimal values depending on the node environment. The proposed JATB algorithm incorporates these optimal transmission power and bit rate based on $N_{vehicle}$. The optimal transmission power and bit rate are compiled into a lookup table. Lookup table is an array that replaces runtime computation with a simpler array indexing operation. With formation of lookup table, the proposed JATB is incorporated into the standard protocol [7] and runs every second to evaluate $N_{vehicle}$. The transmission power and bit rate are adapted according to the transmission conditions or congestion.

4 Output parameters

An extensive simulation was carried out to validate proposed JATB algorithm performance. In this research work, three output parameters were chosen to evaluate the performance

Table 3 Proposed JATB algorithm optimal transmission power and bit rate

$N_{vehicle}$	Transmission Power (mW)	Bit Rate (Mbps)
0–1	18	6
2–6	16	9
7–9	12	12
10–13	8	18
> 13	4	18

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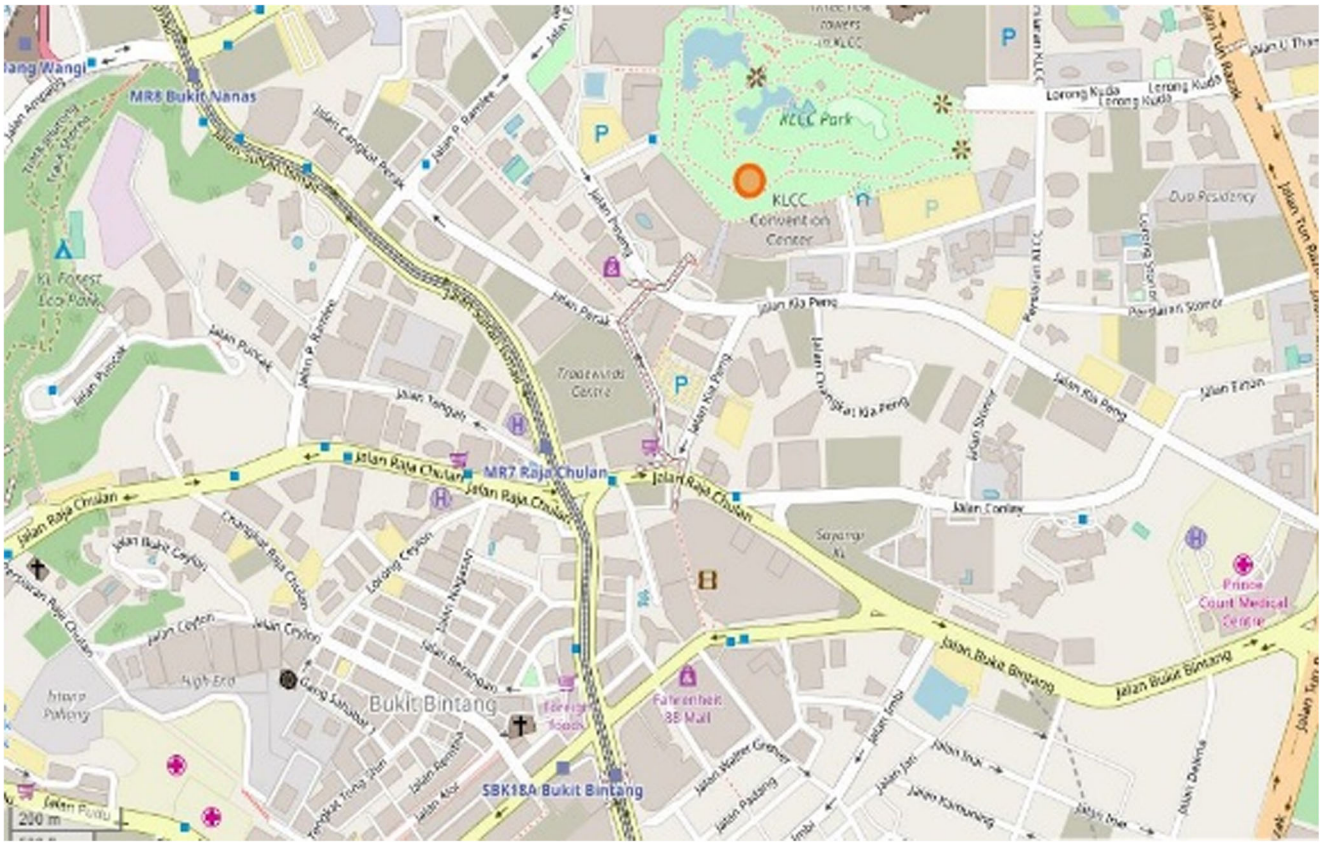

Fig. 4 Map of Kuala Lumpur from openstreetmap.org

of proposed algorithm. Through literatures from [15–18], three output parameters have been chosen to be used in this research work, which are Packet Success Rate, End-to-End delay and Busy Time.

Packet Success Rate indicates the percentage of packets successfully delivered to vehicle. End-to-End delay indicates the time for the packet to reach another vehicle since its creation. Busy Time indicates total time the system is in busy state for sending beacon. Improvement in beacon transmission can be seen through increase in Packet Success Rate, decrease in End-to-End delay and Busy Time.

4.1 Packet success rate

Packet Success Rate is an important output parameter because it calculates the probability of a packet being successfully

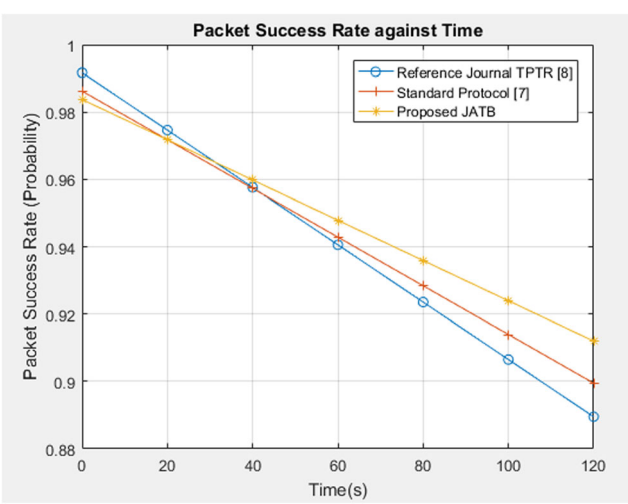

Fig. 5 Comparison of packet success rate (Beacon rate = 0.05 s)

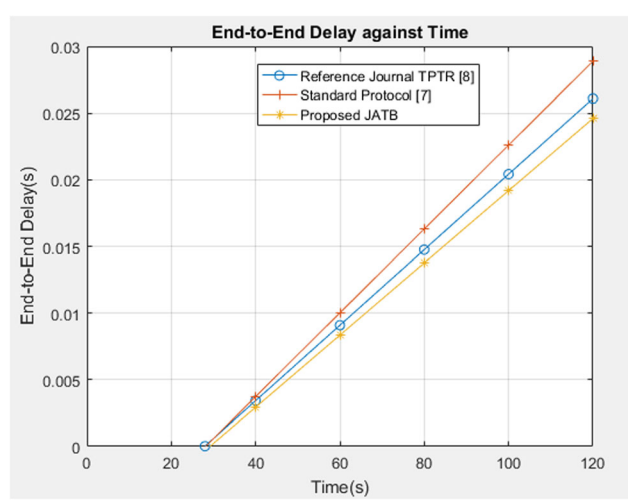

Fig. 6 Comparison of end-to-end delay (Beacon rate = 0.05 s)

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Table 4 Simulation parameter

Simulation Parameters	OMNeT++, SUMO, Veins Values
Map	Kuala Lumpur
Beacon Size	500 bytes
Max Vehicle Number	46
Beacon Rate	0.05, 0.04, 0.03 s
Simulation Time	120 s
Transmission Power	4 – 18 mW
Bit Rate	6–18 Mbps

delivered from transmitter to receiver. It utilizes number of Sent Packet, number of SNIR Lost Packet, number of TXRX Lost packet to calculate Packet Success Rate as shown in Eq. 7.

$$Packet_{SR} = \frac{(N_{SENT} - N_{SNIR} - N_{TXRX})}{N_{SENT}} \tag{7}$$

where N_{SENT} is the number of sent packet, N_{SNIR} is the number of SNIR Lost Packet and N_{TXRX} is the number of TXRX Lost Packet. SNIR Lost Packet is the packet that is not received due to bit error and TXRX Lost Packet is defined as the packet that is not received due to collision as the vehicle is sending while receiving at the same time. High Packet Success Rate indicates high probability of packets that are successfully transmitted across to the receivers, thus a good performing protocol should have high Packet Success Rate.

4.2 End-to-end delay

End-to-end delay is the total time taken since the creation of the packet till the delivery of the message to receiver. The shorter the End-to-End delay the better it is as it signifies faster packet transmission. It is important to keep the delay as short as possible because beacon transmission must be fast to prevent the beacon from becoming irrelevant. End-to-End delay of every packets is summed and the total End-to-End delay is recorded every second. Average total End-to-End delay of nodes is used to evaluate the performance of simulation. Eq. 8 shows the End-to-End delay calculation.

$$Total\ EEdelay_{Average} = \frac{\sum_{n:0}^{n:X}\sum_{p:0}^{p:Y}(end\text{-}to\text{-}end\ delay)}{Total\ number\ of\ nodes} \tag{8}$$

where n is node count, p is packet count.

4.3 Busy time

Busy Time is the total time the node is in busy state. The node is in busy state when it receives or transmits packets. Busy Time must be short to ensure fast packet transmission. Busy Time is recorded every time the node goes into busy state and its average is calculated for comparison.

$$Total\ Busy\ Time_{Average} = \frac{\sum_{n:0}^{n:X}\sum_{p:0}^{p:Y}(Busy\ Time)}{Total\ number\ of\ nodes} \tag{9}$$

where n is node count, p is packet count.

5 Performance comparison

5.1 Benchmarked algorithm

Two algorithms are benchmarked against the proposed JATB algorithm. These algorithms are the standard protocol [7] and TPTR algorithm [8].

1) Standard Protocol Algorithm

Standard protocol [7] is the original protocol set in VANET standard which has fixed transmission power (20 mW) and fixed bit rate (6Mbps).

2) TPTR Algorithm

Reference journal [8] adapts transmission power according to transmission range which is calculated using Eq. 10

$$TR = \min\left\{L(1-K), \sqrt{\frac{L\ln L}{K}} + \alpha L\right\} \tag{10}$$

where α is a constant from traffic flow theory, L is the length of the road segment, $K = AN/TN$ where AN is actual number of vehicles that are preset within the transmission range and TN is the total number of cars that can be present on road with current transmission range. With obtained transmission range (TR), these information is used to form a lookup table. Due to different software used by Danda ea. al, adjustment to the lookup table has to be done. To implement TPTR algorithm

Table 5 Performance comparison (Beacon rate = 0.05 s)

Parameters	Performance Improvement of Proposed JATB (Compared with Standard Protocol [7])	Performance Improvement of the Proposed JATB (Compared with Reference Journal [8])
Packet Success Rate	Higher by 1% (Improved)	Higher by 2% (Improved)
End-to-End delay	Lower by 6% (Improved)	Lower by 4% (Improved)
Busy Time	Lower by 1% (Improved)	Lower by 0.7% (Improved)

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Table 6 Comparison of busy time (Beacon rate = 0.05 s)

Algorithm	Busy Time (s)
Standard Protocol [7]	0.115685 (Highest)
Reference Journal (TPTR) [8]	0.115338 (Medium)
Proposed JATB	0.114500 (Lowest)

[5] into OMNeT++, number of SNIR Lost Packet is used instead of Transmission range. The change is possible due to the linear relationship of Transmission range and number of SNIR Lost Packet. The larger the transmission range the higher the number of SNIR Lost Packet as interference and noise increase (Table 1).

3) The Proposed Joint Adaptation of Transmission power and Bit rate (JATB) algorithm

The proposed JATB algorithm is designed based on the concept of DP. The range of transmission power and bit rate are shown in Table 2. Table 3 shows the selected optimal transmission power and bit rate implemented in OMNeT++.

5.2 Simulation scenarios

For realistic simulation, the simulation is done using real Kuala Lumpur urban map [19] as shown in Fig. 4.

5.3 Result and discussions

From Fig. 5, it can be observed that the proposed JATB algorithm shows higher Packet Success Rate. The packet success rate is 1% higher than the standard protocol [7] and 2% higher than reference journal [8]. From Fig. 6, it can be observed that the proposed JATB shows 4% lower End-to-End delay as compared to TPTR [8] algorithm. The proposed JATB also shows 6% lower End-to-End delay as compared to the standard protocol algorithm [7]. Furthermore, the proposed JATB also shows 0.7% improvement in Busy Time as compared with TPTR algorithm [8] and 1% improvement compared with standard protocol algorithm [7]. All the above results have demonstrated that proposed JATB algorithm outperform both standard protocol [7] and TPTR algorithm [8].

Standard protocol algorithm [7] underperforms when compared with the proposed JATB algorithm. This is because the standard protocol [7] does not include vehicle environment awareness or channel condition measurement, hence keeping its protocol parameter static without any mean of adaptation (Tables 4, 5 and 6).

As for the TPTR [8] algorithm, it does not perform as well as the proposed JATB algorithm because it adapts its transmission power based on only SNIR Lost Packet. SNIR Lost Packet alone is not sufficient to be used as a channel condition estimator because it does not accurately represent the vehicular environment. In addition, reference algorithm [8] only adapts its transmission according to single protocol parameter, transmission power, therefore its transmission efficiency is not as good as the proposed JATB algorithm. This is because the proposed JATB algorithm gives a better accuracy in terms of congestion detection based on two parameters, transmission power and bit rate as compared to a single parameter detection using transmission power in reference algorithm [8].

Complexity comparison between algorithms have also been done as shown in Table 7. We can observe that the complexity of proposed JATB algorithm is 30% less complex when compared with TPTR algorithm. This shows that proposed JATB algorithm utilizes less resource and is more efficient in improving transmission efficiency.

Table 7 Complexity comparison between algorithms with standard protocol as default

Algorithm	Complexity with default as base with 10% factor (in %)
Standard Protocol [7]	–
Reference Journal (TPTR) [8]	82.5%
Proposed JATB	52.5%

6 Conclusion

VANET is an emerging technology that would be deployed onto majority vehicles with the aim to improve transportation system. Ad hoc network between cars should be able to adapt to highly dynamic environment. Therefore, algorithm that makes VANET adaptable is important. In this paper, JATB algorithm is proposed to improve VANET transmission efficiency. The proposed algorithm adapts transmission power and bit rate based on vehicle density. The proposed JATB has been simulated in a realistic urban map or Kuala Lumpur Malaysia. The proposed JATB has shown improvement in Packet Success Rate, End-to-End delay, Busy Time and more efficient with lower complexity as compared with standard protocol [7] and TPTR algorithm [8].

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